Evaluation runs: **pbkv** (a2-store)

generated 2026-11-05 from brain-data 0799b9755 2026-11-05 11:44:26 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): pb: expected conforms; pb-br: expected conforms; control-early-ack-primary: expected conforms; control-early-ack-fwd: expected conforms; silent-put: expected violation; early-ack: expected violation; early-ack: expected conforms (one operation: nothing to reorder or read stale); stale-get: expected violation; stale-get: expected conforms (one operation: nothing to reorder or read stale); sh: expected conforms; misroute: expected violation; misroute: expected conforms (one shard, or one operation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	20	conforms	0.0	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	70	conforms	0.2	4	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,5	252	252	conforms	2.3	4	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,6	924	924	conforms	26.8	13	1 h, 250 GB	924	427350	brain03, 2026-11-02
2,7	3432	3432	conforms	327.1	20	1 h, 250 GB	3432	5891028	brain03, 2026-11-02
3,2	90	90	conforms	0.3	2	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,3	1680	1680	conforms	78.6	13	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,4	34650		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	2520	conforms	184.1	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,3	369600		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4	63063000		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	20	conforms	0.0	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	70	conforms	0.2	4	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,5	252	252	conforms	1.9	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,6	924	924	conforms	31.0	14	1 h, 250 GB	924	427350	brain03, 2026-11-02
2,7	3432	3432	conforms	394.5	19	1 h, 250 GB	3432	5891028	brain03, 2026-11-02
3,2	90	90	conforms	0.2	2	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,3	1680	1680	conforms	75.8	12	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,4	34650		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	2520	conforms	186.7	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,3	369600		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	20	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	70	conforms	0.2	4	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,5	252	252	conforms	2.1	4	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,6	924	924	conforms	26.3	18	1 h, 250 GB	924	427350	brain03, 2026-11-02
2,7	3432	3432	conforms	214.3	25	1 h, 250 GB	3432	5891028	brain03, 2026-11-02
3,2	90	90	conforms	0.2	2	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,3	1680	1680	conforms	55.0	14	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,4	34650		(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	2520	conforms	182.2	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,3	369600		(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4	63063000		(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400		(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	20	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	70	conforms	0.2	4	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,5	252	252	conforms	2.1	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,6	924	924	conforms	26.4	18	1 h, 250 GB	924	427350	brain03, 2026-11-02
2,7	3432	3432	conforms	214.4	25	1 h, 250 GB	3432	5891028	brain03, 2026-11-02
3,2	90	90	conforms	0.2	4	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,3	1680	1680	conforms	62.6	14	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,4	34650		(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	2520	conforms	109.6	15	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,3	369600		(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4	63063000		(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400		(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2,1	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
2,2,2	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
2,2,3	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
2,2,4	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3,1	20	20	conforms	0.0	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,3,2	20	20	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,3,3	20	20	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,3,4	20	20	conforms	0.0	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4,1	70	70	conforms	0.2	2	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,4,2	70	70	conforms	0.2	2	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,4,3	70	70	conforms	0.2	2	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,4,4	70	70	conforms	0.2	2	1 h, 250 GB	70	2485	brain03, 2026-11-02
2,5,1	252	252	conforms	2.2	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,5,2	252	252	conforms	2.0	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,5,3	252	252	conforms	2.1	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
2,5,4	252	252	conforms	2.2	2	1 h, 250 GB	252	31878	brain03, 2026-11-02
3,2,1	90	90	conforms	0.2	4	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,2,2	90	90	conforms	0.2	4	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,2,3	90	90	conforms	0.3	4	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,2,4	90	90	conforms	0.2	4	1 h, 250 GB	90	4095	brain03, 2026-11-02
3,3,1	1680	1680	conforms	63.6	13	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,3,2	1680	1680	conforms	63.3	14	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,3,3	1680	1680	conforms	66.1	13	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,3,4	1680	1680	conforms	66.2	13	1 h, 250 GB	1680	1412040	brain03, 2026-11-02
3,4,1	34650		(1 h)			1 h, 250 GB			brain03, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
4,2,1	2520	2520	conforms	179.1	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,2,2	2520	2520	conforms	173.2	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,2,3	2520	2520	conforms	174.6	12	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,2,4	2520	2520	conforms	172.2	13	1 h, 250 GB	2520	3176460	brain03, 2026-11-02
4,3,1	369600		(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2,1	113400		(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **early-ack**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	10	7	violation	0.0	2	1 h, 250 GB	10	53	brain03, 2026-11-02
2,3	20	54	42	violation	0.1	4	1 h, 250 GB	51	937	brain03, 2026-11-02
2,4	70	300	269	violation	1.1	4	1 h, 250 GB	293	19692	brain03, 2026-11-02
2,5	252	1736	1588	violation	13.3	15	1 h, 250 GB	1663	409767	brain03, 2026-11-02
2,6	924	10102	9719	violation	182.6	40	1 h, 250 GB	9931	9080757	brain03, 2026-11-02
2,7	3432	60106	58212	violation	1896.2	192	1 h, 250 GB	58887	200961061	brain03, 2026-11-02
3,2	90	196	174	violation	0.9	4	1 h, 250 GB	196	16501	brain03, 2026-11-02
3,3	1680	5142	4811	violation	217.0	26	1 h, 250 GB	5073	8262190	brain03, 2026-11-02
3,4	34650			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	4848	4604	violation	389.5	26	1 h, 250 GB	4848	11866342	brain03, 2026-11-02
4,3	369600			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4				(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400			(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **misroute**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2,2	6	6	6	violation	0.0	2	1 h, 250 GB	6	36	brain03, 2026-11-02
2,2,3	6	6	6	violation	0.0	2	1 h, 250 GB	6	36	brain03, 2026-11-02
2,2,4	6	6	6	violation	0.0	2	1 h, 250 GB	6	36	brain03, 2026-11-02
2,3,2	20	20	20	violation	0.0	4	1 h, 250 GB	20	400	brain03, 2026-11-02
2,3,3	20	20	20	violation	0.1	4	1 h, 250 GB	20	400	brain03, 2026-11-02
2,3,4	20	20	20	violation	0.0	2	1 h, 250 GB	20	400	brain03, 2026-11-02
2,4,2	70	70	70	violation	0.3	2	1 h, 250 GB	70	4900	brain03, 2026-11-02
2,4,3	70	70	70	violation	0.3	2	1 h, 250 GB	70	4900	brain03, 2026-11-02
2,4,4	70	70	70	violation	0.3	2	1 h, 250 GB	70	4900	brain03, 2026-11-02
2,5,2	252	252	252	violation	4.2	4	1 h, 250 GB	252	63504	brain03, 2026-11-02
2,5,3	252	252	252	violation	4.2	2	1 h, 250 GB	252	63504	brain03, 2026-11-02
2,5,4	252	252	252	violation	4.2	4	1 h, 250 GB	252	63504	brain03, 2026-11-02
3,2,2	90	90	90	violation	0.5	4	1 h, 250 GB	90	8100	brain03, 2026-11-02
3,2,3	90	90	90	violation	0.5	2	1 h, 250 GB	90	8100	brain03, 2026-11-02
3,2,4	90	90	90	violation	0.5	4	1 h, 250 GB	90	8100	brain03, 2026-11-02
3,3,2	1680	1680	1680	violation	123.3	14	1 h, 250 GB	1680	2822400	brain03, 2026-11-02
3,3,3	1680	1680	1680	violation	131.4	14	1 h, 250 GB	1680	2822400	brain03, 2026-11-02
3,3,4	1680	1680	1680	violation	132.0	14	1 h, 250 GB	1680	2822400	brain03, 2026-11-02
3,4,2	34650			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2,2	2520	2520	2520	violation	343.6	16	1 h, 250 GB	2520	6350400	brain03, 2026-11-02
4,2,3	2520	2520	2520	violation	341.3	18	1 h, 250 GB	2520	6350400	brain03, 2026-11-02
4,2,4	2520	2520	2520	violation	340.7	16	1 h, 250 GB	2520	6350400	brain03, 2026-11-02
4,3,2	369600			(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2,2	113400			(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **silent-put**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	3	3	violation	0.0	4	1 h, 250 GB	3	18	brain03, 2026-11-02
2,3	20	3	3	violation	0.0	4	1 h, 250 GB	3	60	brain03, 2026-11-02
2,4	70	3	3	violation	0.0	2	1 h, 250 GB	3	210	brain03, 2026-11-02
2,5	252	3	3	violation	0.1	4	1 h, 250 GB	3	756	brain03, 2026-11-02
2,6	924	3	3	violation	0.3	2	1 h, 250 GB	3	2772	brain03, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,7	3432	3	3	violation	1.3	4	1 h, 250 GB	3	10296	brain03, 2026-11-02
3,2	90	24	24	violation	0.1	2	1 h, 250 GB	24	2160	brain03, 2026-11-02
3,3	1680	80	80	violation	8.2	4	1 h, 250 GB	80	134400	brain03, 2026-11-02
3,4	34650	280	280	violation	647.7	14	1 h, 250 GB	280	9702000	brain03, 2026-11-02
3,5	756756			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	648	648	violation	94.7	12	1 h, 250 GB	648	1632960	brain03, 2026-11-02
4,3	369600			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4	63063000			(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400			(1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **stale-get**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	6	6	violation	0.0	4	1 h, 250 GB	6	36	brain03, 2026-11-02
2,3	20	20	20	violation	0.0	4	1 h, 250 GB	20	400	brain03, 2026-11-02
2,4	70	70	70	violation	0.3	2	1 h, 250 GB	70	4900	brain03, 2026-11-02
2,5	252	252	252	violation	4.1	2	1 h, 250 GB	252	63504	brain03, 2026-11-02
2,6	924	924	924	violation	48.8	15	1 h, 250 GB	924	853776	brain03, 2026-11-02
2,7	3432	3432	3432	violation	430.5	26	1 h, 250 GB	3432	11778624	brain03, 2026-11-02
3,2	90	90	90	violation	0.5	4	1 h, 250 GB	90	8100	brain03, 2026-11-02
3,3	1680	1680	1680	violation	121.1	14	1 h, 250 GB	1680	2822400	brain03, 2026-11-02
3,4	34650			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	2520	2520	violation	334.0	14	1 h, 250 GB	2520	6350400	brain03, 2026-11-02
4,3	369600			(1 h)			1 h, 250 GB			brain03, 2026-11-02
4,4	63063000			(1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2	113400			(1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): misroute 4 3 3 fifo, misroute 4 3 4 fifo, misroute 4 4 2 fifo, sh 4 4 3 fifo, misroute 4 4 3 fifo, misroute 4 4 4 fifo, misroute 4 5 2 fifo, misroute 4 5 3 fifo, misroute 4 5 4 fifo, misroute 5 2 3 fifo, misroute 5 2 4 fifo, misroute 5 3 2 fifo, misroute 5 3 3 fifo, misroute 5 3 4 fifo, misroute 5 4 2 fifo, misroute 5 4 3 fifo, misroute 5 4 4 fifo, misroute 5 5 2 fifo, misroute 5 5 3 fifo, misroute 5 5 4 fifo.

skipped: a smaller size had reached its limit: sh 3 4 2 fifo, misroute 3 4 3 fifo, sh 3 4 3 fifo, sh 3 4 4 fifo, misroute 3 4 4 fifo, pb 3 5 fifo, stale-get 3 5 fifo, pb-br 3 5 fifo, early-ack 3 5 fifo, control-early-ack-primary 3 5 fifo, control-early-ack-fwd 3 5 fifo, sh 3 5 1 fifo, sh 3 5 2 fifo, misroute 3 5 2 fifo, misroute 3 5 3 fifo, sh 3 5 3 fifo, sh 3 5 4 fifo, misroute 3 5 4 fifo, silent-put 3 6 fifo, early-ack 3 6 fifo, control-early-ack-primary 3 6 fifo, pb 3 6 fifo, stale-get 3 6 fifo, control-early-ack-fwd 3 6 fifo, … (198 instances in all).

stateful

stateful: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	20	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	924	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3432	conforms	0.6	4	1 h, 250 GB	brain02, 2026-11-02
3,2	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	1680	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	34650	conforms	3.2	6	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	756756	conforms	32.2	247	1 h, 250 GB	brain02, 2026-11-02
3,6			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	2520	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	369600	conforms	30.2	114	1 h, 250 GB	brain02, 2026-11-02
4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	113400	conforms	18.1	36	1 h, 250 GB	brain02, 2026-11-02
5,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	7484400	conforms	1580.0	1727	1 h, 250 GB	brain02, 2026-11-02
6,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	924	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3432	conforms	0.5	4	1 h, 250 GB	brain02, 2026-11-02
3,2	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	1680	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	34650	conforms	3.0	6	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	756756	conforms	45.3	245	1 h, 250 GB	brain02, 2026-11-02
3,6			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	2520	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	369600	conforms	34.1	105	1 h, 250 GB	brain02, 2026-11-02
4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	113400	conforms	15.9	36	1 h, 250 GB	brain02, 2026-11-02
5,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	7484400	conforms	1325.3	1727	1 h, 250 GB	brain02, 2026-11-02
6,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	924	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3432	conforms	0.5	6	1 h, 250 GB	brain02, 2026-11-02
3,2	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	1680	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	34650	conforms	1.8	16	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	756756	conforms	21.8	248	1 h, 250 GB	brain02, 2026-11-02
3,6			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	2520	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	369600	conforms	23.6	118	1 h, 250 GB	brain02, 2026-11-02
4,4			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	113400	conforms	14.5	38	1 h, 250 GB	brain02, 2026-11-02
5,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	7484400	conforms	983.4	1731	1 h, 250 GB	brain02, 2026-11-02
6,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	924	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3432	conforms	0.4	6	1 h, 250 GB	brain02, 2026-11-02
3,2	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	1680	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	34650	conforms	1.8	20	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	756756	conforms	20.8	250	1 h, 250 GB	brain02, 2026-11-02
3,6			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	2520	conforms	0.4	2	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	369600	conforms	12.0	121	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
4,4			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	113400	conforms	8.0	24	1 h, 250 GB	brain02, 2026-11-02
5,3			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	7484400	conforms	413.9	1738	1 h, 250 GB	brain02, 2026-11-02
6,3			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2,1	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,2,2	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,2,3	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,2,4	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3,1	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,2	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,3	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,4	20	20	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,1	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,2	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,3	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,4	70	70	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5,1	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,5,2	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,5,3	252	252	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,5,4	252	252	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,2,1	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,2,2	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,2,3	90	90	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,2,4	90	90	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3,1	1680	1680	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-02
3,3,2	1680	1680	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-02
3,3,3	1680	1680	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-02
3,3,4	1680	1680	conforms	0.3	4	1 h, 250 GB	brain02, 2026-11-02
3,4,1	34650	34650	conforms	1.9	12	1 h, 250 GB	brain02, 2026-11-02
3,4,2	34650	34650	conforms	2.0	16	1 h, 250 GB	brain02, 2026-11-02
3,4,3	34650	34650	conforms	2.1	14	1 h, 250 GB	brain02, 2026-11-02
3,4,4	34650	34650	conforms	2.0	20	1 h, 250 GB	brain02, 2026-11-02
3,5,1	756756	756756	conforms	22.0	248	1 h, 250 GB	brain02, 2026-11-02
3,5,2	756756	756756	conforms	22.5	249	1 h, 250 GB	brain02, 2026-11-02
3,5,3	756756	756756	conforms	23.1	249	1 h, 250 GB	brain02, 2026-11-02
3,5,4	756756	756756	conforms	23.5	250	1 h, 250 GB	brain02, 2026-11-02
4,2,1	2520	2520	conforms	0.5	2	1 h, 250 GB	brain02, 2026-11-02
4,2,2	2520	2520	conforms	0.6	4	1 h, 250 GB	brain02, 2026-11-02
4,2,3	2520	2520	conforms	0.5	2	1 h, 250 GB	brain02, 2026-11-02
4,2,4	2520	2520	conforms	0.5	4	1 h, 250 GB	brain02, 2026-11-02
4,3,1	369600	369600	conforms	15.5	119	1 h, 250 GB	brain02, 2026-11-02
4,3,2	369600	369600	conforms	15.7	120	1 h, 250 GB	brain02, 2026-11-02
4,3,3	369600	369600	conforms	15.9	121	1 h, 250 GB	brain02, 2026-11-02
4,3,4	369600	369600	conforms	15.9	121	1 h, 250 GB	brain02, 2026-11-02
4,4,1			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2,1	113400	113400	conforms	15.2	39	1 h, 250 GB	brain02, 2026-11-02
5,2,2	113400	113400	conforms	15.3	39	1 h, 250 GB	brain02, 2026-11-02
5,2,3	113400	113400	conforms	14.9	40	1 h, 250 GB	brain02, 2026-11-02
5,2,4	113400	113400	conforms	15.5	40	1 h, 250 GB	brain02, 2026-11-02
5,3,1			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **early-ack**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	10	7	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	54	42	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,4	70	300	269	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	1736	1588	violation	0.2	8	1 h, 250 GB	brain02, 2026-11-02
2,6	924	10102	9719	violation	0.6	46	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	60106	58212	violation	2.1	242	1 h, 250 GB	brain02, 2026-11-02
3,2	90	196	174	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	5142	4811	violation	0.5	28	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	224810	221544	violation	7.9	1406	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	6847694	6784495	violation	777.9	52912	1 h, 250 GB	brain02, 2026-11-02
3,6				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	4848	4604	violation	0.6	30	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	1320692	1294900	violation	54.1	8495	1 h, 250 GB	brain02, 2026-11-02
4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	258768	254220	violation	14.7	1854	1 h, 250 GB	brain02, 2026-11-02
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	15512112	15403284	violation	1918.5	130648	1 h, 250 GB	brain02, 2026-11-02
6,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **misroute**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2,2	6	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,2,3	6	6	6	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,2,4	6	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,2	20	20	20	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,3	20	20	20	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3,4	20	20	20	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4,2	70	70	70	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,3	70	70	70	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4,4	70	70	70	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5,2	252	252	252	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5,3	252	252	252	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,5,4	252	252	252	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
3,2,2	90	90	90	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,2,3	90	90	90	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,2,4	90	90	90	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3,2	1680	1680	1680	violation	0.3	12	1 h, 250 GB	brain02, 2026-11-02
3,3,3	1680	1680	1680	violation	0.3	12	1 h, 250 GB	brain02, 2026-11-02
3,3,4	1680	1680	1680	violation	0.3	12	1 h, 250 GB	brain02, 2026-11-02
3,4,2	34650	34650	34650	violation	2.7	291	1 h, 250 GB	brain02, 2026-11-02
3,4,3	34650	34650	34650	violation	2.8	292	1 h, 250 GB	brain02, 2026-11-02
3,4,4	34650	34650	34650	violation	2.9	292	1 h, 250 GB	brain02, 2026-11-02
3,5,2	756756	756756	756756	violation	46.0	8485	1 h, 250 GB	brain02, 2026-11-02
3,5,3	756756	756756	756756	violation	46.5	8494	1 h, 250 GB	brain02, 2026-11-02
3,5,4	756756	756756	756756	violation	46.6	8487	1 h, 250 GB	brain02, 2026-11-02
4,2,2	2520	2520	2520	violation	0.5	16	1 h, 250 GB	brain02, 2026-11-02
4,2,3	2520	2520	2520	violation	0.5	18	1 h, 250 GB	brain02, 2026-11-02
4,2,4	2520	2520	2520	violation	0.6	18	1 h, 250 GB	brain02, 2026-11-02
4,3,2	369600	369600	369600	violation	24.0	3106	1 h, 250 GB	brain02, 2026-11-02
4,3,3	369600	369600	369600	violation	24.5	3102	1 h, 250 GB	brain02, 2026-11-02
4,3,4	369600	369600	369600	violation	25.2	3110	1 h, 250 GB	brain02, 2026-11-02
4,4,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2,2	113400	113400	113400	violation	17.0	883	1 h, 250 GB	brain02, 2026-11-02
5,2,3	113400	113400	113400	violation	18.0	887	1 h, 250 GB	brain02, 2026-11-02
5,2,4	113400	113400	113400	violation	17.8	882	1 h, 250 GB	brain02, 2026-11-02
5,3,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **silent-put**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	3	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	3	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,4	70	3	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	3	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	3	3	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3	3	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
3,2	90	24	24	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	80	80	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	280	280	violation	0.7	8	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	1008	1008	violation	9.9	172	1 h, 250 GB	brain02, 2026-11-02
3,6	17153136	3696	3696	violation	2034.2	2893	1 h, 250 GB	brain02, 2026-11-02
3,7				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	648	648	violation	0.3	4	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	12040	12040	violation	7.5	163	1 h, 250 GB	brain02, 2026-11-02
4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	19980	19980	violation	7.8	168	1 h, 250 GB	brain02, 2026-11-02
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	1360800	1360800	violation	392.6	12853	1 h, 250 GB	brain02, 2026-11-02
6,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **stale-get**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	20	20	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	70	70	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,5	252	252	252	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,6	924	924	924	violation	0.2	6	1 h, 250 GB	brain02, 2026-11-02
2,7	3432	3432	3432	violation	0.5	34	1 h, 250 GB	brain02, 2026-11-02
3,2	90	90	90	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	1680	1680	violation	0.4	14	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	34650	34650	violation	2.6	289	1 h, 250 GB	brain02, 2026-11-02
3,5	756756	756756	756756	violation	45.2	8492	1 h, 250 GB	brain02, 2026-11-02
3,6				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	2520	2520	violation	0.5	16	1 h, 250 GB	brain02, 2026-11-02
4,3	369600	369600	369600	violation	32.5	3112	1 h, 250 GB	brain02, 2026-11-02
4,4				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	113400	113400	violation	16.6	879	1 h, 250 GB	brain02, 2026-11-02
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
6,2	7484400	7484400	7484400	violation	1306.8	66779	1 h, 250 GB	brain02, 2026-11-02
6,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
7,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): pb 6 2 fifo, misroute 6 6 4 fifo, sh 6 6 4 fifo.

skipped: a smaller size had reached its limit: control-early-ack-fwd 3 7 fifo, stale-get 3 7 fifo, early-ack 3 7 fifo, pb 3 7 fifo, control-early-ack-primary 3 7 fifo, pb-br 3 7 fifo, sh 4 4 2 fifo, misroute 4 4 3 fifo, sh 4 4 3 fifo, misroute 4 4 4 fifo, sh 4 4 4 fifo, control-early-ack-primary 4 5 fifo, stale-get 4 5 fifo, pb-br 4 5 fifo, control-early-ack-fwd 4 5 fifo, silent-put 4 5 fifo, pb 4 5 fifo, early-ack 4 5 fifo, sh 4 5 1 fifo, misroute 4 5 2 fifo, sh 4 5 2 fifo, sh 4 5 3 fifo, misroute 4 5 3 fifo, misroute 4 5 4 fifo, ... (149 instances in all).

offline

offline: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	1680	conforms	0.5	20	10 min, 16 GB	0.1	0.4	0.1	in-memory	brain01, 2026-11-02
4,4	63063000	63063000	oom (250 GB)			1 h, 250 GB	1606.5	2847.2		in-memory	brain01, 2026-11-02

offline: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	1680	conforms	0.4	20	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02

offline: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	1680	conforms	0.3	20	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain01, 2026-11-02
4,4	63063000	63063000	(1 h)			1 h, 250 GB	1606.5	1980.4		merge	brain01, 2026-11-03

offline: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain01, 2026-11-02
3,3	1680	1680	conforms	0.4	20	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
4,4	63063000	63063000	oom (250 GB)			1 h, 250 GB	1606.5	2001.2		in-memory	brain01, 2026-11-02

offline: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,1	6	6	conforms	0.3	4	10 min, 16 GB	0.2	0.3	0.0	in-memory	brain01, 2026-11-02
2,2,2	6	6	conforms	0.2	4	10 min, 16 GB	0.2	0.0	0.0	in-memory	brain01, 2026-11-02
3,3,1	1680	1680	conforms	0.4	20	10 min, 16 GB	0.3	0.2	0.1	in-memory	brain01, 2026-11-02
3,3,2	1680	1680	conforms	0.5	20	10 min, 16 GB	0.4	0.3	0.1	in-memory	brain01, 2026-11-02

offline: pair **early-ack**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	10	7	violation	0.3	4	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain01, 2026-11-02
3,3	1680	5142	4094	violation	0.7	38	10 min, 16 GB	0.1	0.5	0.2	in-memory	brain01, 2026-11-02

offline: pair **misroute**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,2	6	6	6	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
3,3,2	1680	1680	1680	violation	0.5	22	10 min, 16 GB	0.4	0.2	0.1	in-memory	brain01, 2026-11-02

offline: pair **silent-put**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	3	3	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,3	1680	80	80	violation	0.2	12	10 min, 16 GB	0.1	0.0	0.1	in-memory	brain01, 2026-11-02
4,4	63063000	247800		oom (250 GB)			1 h, 250 GB	1606.5	12.8		in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run

offline: pair **stale-get**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	6	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,3	1680	1680	1680	violation	0.3	22	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain01, 2026-11-02
4,4	63063000	63063000		oom (250 GB)			1 h, 250 GB	1606.5	1945.6		in-memory	brain01, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): misroute 2 2 3 fifo, sh 2 2 3 fifo, misroute 2 2 4 fifo, sh 2 2 4 fifo, early-ack 2 3 fifo, silent-put 2 3 fifo, stale-get 2 3 fifo, control-early-ack-fwd 2 3 fifo, control-early-ack-primary 2 3 fifo, pb 2 3 fifo, pb-br 2 3 fifo, sh 2 3 1 fifo, sh 2 3 2 fifo, misroute 2 3 2 fifo, sh 2 3 3 fifo, misroute 2 3 3 fifo, misroute 2 3 4 fifo, sh 2 3 4 fifo, early-ack 2 4 fifo, control-early-ack-primary 2 4 fifo, pb 2 4 fifo, pb-br 2 4 fifo, control-early-ack-fwd 2 4 fifo, silent-put 2 4 fifo, … (342 instances in all).

skipped: a smaller size had reached its limit: early-ack 4 4 fifo, control-early-ack-primary 4 4 fifo.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	4.4	162	10 min, 16 GB	4.3	0.2	0.1	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	13.4	5076	10 min, 16 GB	13.3	0.4	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.9	153	10 min, 16 GB	3.8	0.2	0.1	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	10.8	4504	10 min, 16 GB	10.7	0.2	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **early-ack**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	10	10	conforms	4.6	167	10 min, 250 GB	4.6	0.3	0.0	merge	brain02, 2026-11-03
3,3	5142	5142	conforms	20.7	3046	10 min, 250 GB	20.4	0.7	0.3	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **misroute**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,2	6	6	conforms	4.1	158	10 min, 16 GB	4.1	0.2	0.0	in-memory	brain02, 2026-11-03
3,3,2	1680	1680	conforms	19.0	4369	10 min, 16 GB	18.9	0.9	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	4.5	152	10 min, 250 GB	4.5	0.3	0.0	merge	brain02, 2026-11-03
3,3	1680	1680	conforms	12.9	2617	10 min, 250 GB	12.7	0.3	0.2	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.4	156	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	11.4	4250	10 min, 16 GB	11.3	0.3	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,1	6	6	conforms	6.1	166	10 min, 16 GB	6.1	0.2	0.0	in-memory	brain02, 2026-11-03
2,2,2	6	6	conforms	4.1	153	10 min, 16 GB	4.1	0.2	0.0	in-memory	brain02, 2026-11-03
3,3,1	1680	1680	conforms	12.8	4715	10 min, 16 GB	12.7	0.3	0.1	in-memory	brain02, 2026-11-03
3,3,2	1680	1680	conforms	12.8	4418	10 min, 16 GB	12.7	0.3	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **silent-put**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	3	3	conforms	3.6	147	10 min, 16 GB	3.6	0.2	0.0	in-memory	brain02, 2026-11-03
3,3	80	80	conforms	4.5	156	10 min, 16 GB	4.5	0.2	0.0	in-memory	brain02, 2026-11-03
4,4	247800	247800	conforms	1246.4	20150	10 min, 16 GB	1215.0	13.5	31.4	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **stale-get**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.8	157	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	10.5	2668	10 min, 16 GB	10.4	0.3	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). skipped: a smaller size had reached its limit: pb 4 4 fifo, early-ack 4 4 fifo, control-early-ack-primary 4 4 fifo.
refused (an input missing at run time): stale-get 4 4 fifo, control-early-ack-fwd 4 4 fifo, pb-br 4 4 fifo.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	4.4	162	10 min, 16 GB	0.2	4.3	0.1	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	13.4	5076	10 min, 16 GB	0.4	13.3	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.9	153	10 min, 16 GB	0.2	3.8	0.1	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	10.8	4504	10 min, 16 GB	0.2	10.7	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **early-ack**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	10	10	conforms	4.6	167	10 min, 250 GB	0.3	4.6	0.0	merge	brain02, 2026-11-03
3,3	5142	5142	conforms	20.7	3046	10 min, 250 GB	0.7	20.4	0.3	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **misroute**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,2	6	6	conforms	4.1	158	10 min, 16 GB	0.2	4.1	0.0	in-memory	brain02, 2026-11-03
3,3,2	1680	1680	conforms	19.0	4369	10 min, 16 GB	0.9	18.9	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	4.5	152	10 min, 250 GB	0.3	4.5	0.0	merge	brain02, 2026-11-03
3,3	1680	1680	conforms	12.9	2617	10 min, 250 GB	0.3	12.7	0.1	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.4	156	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	11.4	4250	10 min, 16 GB	0.3	11.3	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,1	6	6	conforms	6.1	166	10 min, 16 GB	0.2	6.1	0.0	in-memory	brain02, 2026-11-03
2,2,2	6	6	conforms	4.1	153	10 min, 16 GB	0.2	4.1	0.0	in-memory	brain02, 2026-11-03
3,3,1	1680	1680	conforms	12.8	4715	10 min, 16 GB	0.3	12.7	0.1	in-memory	brain02, 2026-11-03
3,3,2	1680	1680	conforms	12.8	4418	10 min, 16 GB	0.3	12.7	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **silent-put**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	3	3	conforms	3.6	147	10 min, 16 GB	0.2	3.6	0.0	in-memory	brain02, 2026-11-03
3,3	80	80	conforms	4.5	156	10 min, 16 GB	0.2	4.5	0.0	in-memory	brain02, 2026-11-03
4,4	247800	247800	conforms	1246.5	20150	10 min, 16 GB	13.5	1215.0	31.5	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **stale-get**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	6	conforms	3.8	157	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain02, 2026-11-03
3,3	1680	1680	conforms	10.5	2668	10 min, 16 GB	0.3	10.4	0.1	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: early-ack 4 4 fifo, control-early-ack-fwd 4 4 fifo, pb 4 4 fifo, stale-get 4 4 fifo, pb-br 4 4 fifo, control-early-ack-primary 4 4 fifo.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	3.6	174	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	13.2	4868	1 h, 250 GB		brain02, 2026-11-03
4,4	63063000		oom (250 GB)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	3.9	170	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	12.2	4723	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **early-ack**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	10	10	conforms	4.8	154	1 h, 250 GB		brain02, 2026-11-03
3,3	5142	5142	conforms	26.8	4882	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **misroute**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	6	6	conforms	3.4	164	1 h, 250 GB		brain02, 2026-11-03
3,3,2	1680	1680	conforms	11.7	4832	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **pb**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	3.9	160	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	14.1	3106	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **pb-br**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	3.4	158	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	13.3	4704	1 h, 250 GB		brain02, 2026-11-03
4,4	63063000		oom (250 GB)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **sh**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,1	6	6	conforms	3.6	166	1 h, 250 GB		brain02, 2026-11-03
2,2,2	6	6	conforms	3.4	159	1 h, 250 GB		brain02, 2026-11-03
3,3,1	1680	1680	conforms	12.5	4762	1 h, 250 GB		brain02, 2026-11-03
3,3,2	1680	1680	conforms	12.3	4889	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **silent-put**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	3	3	conforms	3.7	158	1 h, 250 GB		brain02, 2026-11-03
3,3	80	80	conforms	5.0	180	1 h, 250 GB		brain02, 2026-11-03
4,4	247800	247800	conforms	1211.5	58383	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **stale-get**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	3.3	160	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	11.3	4883	1 h, 250 GB		brain02, 2026-11-03
4,4	63063000		oom (250 GB)			1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). skipped: a smaller size had reached its limit: pb 4 4 fifo, early-ack 4 4 fifo, control-early-ack-primary 4 4 fifo.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-early-ack-fwd**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	0.4	14	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-early-ack-primary**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	0.3	12	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair early-ack, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	10	10	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	5142	5142	conforms	0.6	32	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair misroute, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3,2	1680	1680	conforms	0.3	14	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair pb, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	0.1	4	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	0.3	14	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair pb-br, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	0.3	14	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair sh, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,1	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2,2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3,1	1680	1680	conforms	0.3	14	1 h, 250 GB		brain02, 2026-11-03
3,3,2	1680	1680	conforms	0.4	14	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair silent-put, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	3	3	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	80	80	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4,4	247800	247800	conforms	31.8	2646	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair stale-get, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,3	1680	1680	conforms	0.3	14	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). skipped: a smaller size had reached its limit: early-ack 4 4 fifo, control-early-ack-fwd 4 4 fifo, pb 4 4 fifo, stale-get 4 4 fifo, pb-br 4 4 fifo, control-early-ack-primary 4 4 fifo.